Shubhankar Mohakud

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OVERVIEW

Computer Science undergraduate at KIIT, DU, with a strong foundation in machine learning, distributed systems, and high-performance computing. Passionate about building production-ready pipelines that bridge scientific computing with real-world impact — from anomaly detection at LHC scale to intelligent job scheduling simulations.

EDUCATION

Kalinga Institute of Industrial Technology, DU — CGPA 8.5
B.Tech · 2023 - 2027

Bal Bharati Public School, Noida — 92.6%
CBSE Class XII · 2022

Bal Bharati Public School, Noida — 95.0%
CBSE Class X · 2020

TECHNICAL SKILLS

Languages	C / C++, Java, Python, JavaScript, MATLAB
Frameworks	Node.js, Express.js, Scikit-learn, NumPy, Pandas
Databases	PostgreSQL, Oracle, SQL
Tools	Git, Postman, Linux

PROJECTS

01 Scalable Anomaly Detection on LHC-Style Event Data

ML · PyTorch · Scikit-learn

- Unsupervised pipeline using Variational Autoencoders & Isolation Forests — 92% detection accuracy
- Distributed processing with Dask across 10M+ events; 8× faster than serial execution
- Physics-informed metrics (AUC-ROC, signal efficiency) under varying noise and data volumes

02 Parallel Data Processing Framework for Scientific Computing

HPC · Python · Multiprocessing

- High-performance pipeline for ROOT-like columnar datasets — 5-6× speedup on 8-core systems
- Benchmarked multiprocessing, threading, and joblib strategies; 80-87% parallel efficiency at 4 workers
- Profiling tools measuring throughput and execution time across 1M+ events with minimal overhead

03 Intelligent Job Scheduling Simulator for Distributed Computing

Distributed Systems · Python · Matplotlib

- Discrete-event simulation across heterogeneous nodes with realistic fault tolerance modeling
- FIFO, priority-based, adaptive, and SJF algorithms — adaptive scheduling yields 35% throughput gain
- Comprehensive performance reports (fairness, fault tolerance) with Matplotlib visualizations

ACHIEVEMENTS

- ★ Ranked 1st among 500+ teams in Smart India Hackathon (SIH) internal rounds
- ★ 7th rank among 200+ students — CBSE Class X board examinations (95%)
- ★ 10th rank among 100+ students — CBSE Class XII board examinations (92.6%)